CAT 320: Safety & Maintenance

Daily

CATSENSE CAT

● 2 ● 3 ● 3 ● 0

Inspection Number	012071EE-E	Work Order	WO-012071
Serial Number	ZAR00512	Completed On	3/1/2026, 1:45:22 AM
Make	CATERPILLAR	Inspector	Field Officer 01
Model	CAT 320	PDF Generated On	3/1/2026
Equipment Family	Excavator	Location	East Peoria, IL 61611
Asset ID	M1		
SMU	575.3 Hours		

OVERALL STATUS
CRITICAL

8 checks
AI-analyzed

Detailed Report

8 items

FROM THE GROUND
Bucket, GET

NEEDS ATTENTION

EVIDENCE PHOTOS (1)

The hydraulic excavator's bucket exhibits significant structural deformation and extensive surface corrosion. The Ground Engaging Tools (GET), specifically the bucket teeth, show excessive wear and blunting. The overall condition suggests a need for maintenance to restore optimal performance and prevent further structural degradation.

FINDINGS

MED **Bucket Structure**

Significant deformation and compromise of the bucket shell, particularly on the bottom and side plates. Extensive surface corrosion and paint loss are also present.

Evidence: Visible warping and irregular shape of the bucket, heavy rust, and widespread paint peeling across the bucket body.

Action: Conduct a detailed structural inspection of the bucket for cracks, material thinning, and weld integrity. Based on findings, consider bucket repair or replacement to restore structural integrity and performance.

MED **Ground Engaging Tools (GET) - Bucket Teeth**

Excessive wear and blunting observed on multiple bucket teeth, indicating reduced digging efficiency and increased load on the machine.

Evidence: Visibly worn-down and rounded profiles of the bucket teeth, lacking sharp cutting edges.

Action: Replace all excessively worn bucket teeth. Inspect the adapters for wear and replace if necessary to ensure proper fitment and retention of new teeth.

FOLLOW-UP QUESTIONS

1. Please provide closer images of the bucket welds, especially in areas of deformation, to check for any hidden cracks.
2. Can the field inspector confirm the integrity and presence of all retainer pins for the Ground Engaging Tools?
3. Are there any operational reports indicating reduced digging efficiency, increased fuel consumption, or unusual vibrations when using the bucket?

FROM THE GROUND

Hydraulic Oil

OK

EVIDENCE PHOTOS (1)

The provided image displays a fluid level sight glass, which appears to indicate an acceptable fluid level between the specified markers. However, the image does not contain visual evidence of the boom, stick, or bucket hydraulic cylinders, their rods, wiper seals, or hydraulic lines. Therefore, a comprehensive assessment of these specific components for scoring, pitting, leaks, or damage cannot be performed based on the given visual data.

FOLLOW-UP QUESTIONS

1. Please provide clear visual evidence of the boom, stick, and bucket hydraulic cylinders, including their rods, wiper seals, and hydraulic lines, to enable a comprehensive inspection for scoring, pitting, leaks, and damage.

FROM THE GROUND

Sticks

CRITICAL

EVIDENCE PHOTOS (1)

The hydraulic system exhibits severe degradation of multiple hydraulic hoses, including one with exposed reinforcement. There is also evidence of hydraulic fluid leaks around fittings and on the main hydraulic components.

FINDINGS

MED **Hydraulic Hoses**

Severe cracking and degradation of the outer rubber cover on multiple hydraulic hoses.

Evidence: Visible deep, widespread cracks on the outer surface of several hydraulic hoses.

Action: Schedule replacement of all hydraulic hoses exhibiting severe outer cover degradation.

HIGH **Hydraulic Hose**

Outer cover completely failed, exposing the braided reinforcement layer.

Evidence: Clearly visible braided reinforcement material through a large section where the outer rubber cover is missing.

Action: Immediate replacement of the compromised hydraulic hose to prevent catastrophic failure.

HIGH **Hydraulic System Fittings and Components**

Evidence of hydraulic fluid leakage or significant seepage.

Evidence: Oily residue and wetness observed on and around multiple hydraulic fittings and on the surface of the yellow hydraulic control valves/pumps.

Action: Thoroughly inspect all fittings, seals, and hose crimps to identify and repair the source of the hydraulic fluid leaks. Clean the affected area after repair.

FOLLOW-UP QUESTIONS

1. What is the current operational hour meter reading for this excavator?
2. Has the hydraulic fluid level been checked recently, and if so, was it low?
3. Are there any reported performance issues with the hydraulic system, such as sluggish operation or loss of power?
4. Can the inspector confirm if the observed fluid is actively leaking or residual from previous events?

ENGINE COMPARTMENT

All Hoses

CRITICAL

EVIDENCE PHOTOS (1)

The inspection reveals critical degradation of hydraulic system components, including extensive dry rot and cracking on multiple hydraulic hoses, and an active hydraulic fluid leak originating from a pump/valve assembly and a hose. Additionally, several electrical harnesses exhibit severe abrasion and wear, with one showing fluid contamination. These conditions pose significant risks to operational safety and equipment integrity.

FINDINGS

HIGH **Hydraulic Hoses**

Extensive dry rot and cracking observed on the outer cover of multiple hydraulic hoses, indicating severe material degradation and potential for imminent failure.

Evidence: Visual inspection of multiple gray-colored hydraulic hoses showing deep, widespread cracking patterns on their exterior, as highlighted in 'CRACKS' crops.

Action: Immediate replacement of all affected hydraulic hoses. Pressure testing of the hydraulic system after replacement to confirm integrity.

MED **Electrical Harnesses**

Outer braiding of multiple electrical harnesses is severely abraded and worn, exposing internal wiring insulation. One harness shows signs of fluid contamination/residue.

Evidence: Visual inspection of beige-braided electrical harnesses with significant fraying and material loss, revealing underlying wires, as highlighted in 'WEAR SPOTS' crops. Dark, possibly oily, residue observed on one worn section.

Action: Inspect exposed wiring for damage, repair or replace damaged sections of the harness, and secure all harnesses to prevent further abrasion. Investigate the source of the fluid contamination.

HIGH **Hydraulic Pump/Valve Assembly and Hydraulic Hose**

Evidence of active hydraulic fluid leakage around the base of a hydraulic pump or valve assembly, and a distinct drip on a hydraulic hose, indicating a seal failure, loose fitting, or hose compromise.

Evidence: Visible pooling of iridescent fluid around the bolted base of a yellow hydraulic component, and a clear droplet of fluid on the underside of a black hydraulic hose, as highlighted in 'LEAKS' crop.

Action: Identify the exact source of the leak (fitting, seal, hose), repair or replace the faulty component, and replenish hydraulic fluid to the correct level. Clean the affected area thoroughly.

FOLLOW-UP QUESTIONS

1. What is the operational history of this excavator, specifically regarding hours and recent maintenance?
2. Can the inspector confirm the type of fluid leaking (e.g., hydraulic oil, engine oil)?
3. Are there any noticeable performance issues or warning lights related to the hydraulic system?

ENGINE COMPARTMENT

Overall Engine

OK

EVIDENCE PHOTOS (1)

The engine compartment exhibits typical operational dust and grime. No active oil leaks from valve covers or the oil pan, water in the primary fuel filter/water separator bowl, or chafed fuel lines are visually evident from the provided image.

FOLLOW-UP QUESTIONS

1. Can a clearer image or video be provided of the primary fuel filter/water separator bowl to confirm the absence of water?
2. Can a closer inspection be performed on all fuel lines, particularly in areas prone to movement or contact, to definitively rule out chafing?

3. Are there any recent operational reports or historical data indicating abnormal oil consumption or fuel system alerts?

INSIDE THE CAB

Controls

NEEDS ATTENTION

EVIDENCE PHOTOS (1)

The monitor display is observed to be blank, which prevents the verification of active diagnostic codes or warning lights. No other critical warning indicators are illuminated on the dashboard, and the visible control panel components appear to be in good physical condition. The operational status of joysticks, travel pedals, travel alarm, and horn could not be assessed from the provided data.

FINDINGS

MED **Monitor Display**

Monitor display is blank, preventing verification of active diagnostic codes or warning lights.

Evidence: Visual observation of a non-illuminated or blank monitor screen.

Action: Verify the excavator's power status. If powered on, troubleshoot the monitor display system for functionality. If off, power on the machine and confirm display operation.

FOLLOW-UP QUESTIONS

1. Is the excavator currently powered on?
2. If powered on, does the monitor display remain blank, or does it cycle through a startup sequence?

INSIDE THE CAB

Master Switch

NEEDS ATTENTION

EVIDENCE PHOTOS (1)

The provided visual evidence does not depict the specified components for inspection, namely the Cab Glass, Mirrors, or Falling Object Protective Structure (FOPS). Therefore, a condition assessment for these items cannot be performed based on the available image.

FOLLOW-UP QUESTIONS

1. Please provide additional images that clearly show the Cab Glass, Mirrors, and Falling Object Protective Structure (FOPS) to enable a complete and accurate assessment.

INSIDE THE CAB

Seat

OK

EVIDENCE PHOTOS (1)

The provided image does not contain any visible components related to the seat, seat belt, fire extinguisher, or safety lock lever of a CAT Hydraulic Excavator. Therefore, no assessment of these components can be made.

FOLLOW-UP QUESTIONS

1. Please provide images that clearly show the seat belt for fraying or latch failure, the fire extinguisher for presence and charge status, and the safety lock lever for operation.